

D Test Plan

This document lays out how someone is able to ensure the BMS is fully operational.

D.1 Top Down Testing

D.1.1 Purpose

To test the entire system against its requirements by performing a top-down test.

D.1.2 System Equipment

Equipment Needed

- System Equipment
 - Battery Monitoring Module
 - Battery Management Unit (BMU)
 - Current SHUNT Module
 - 1.5 miliOhm SHUNT Resistor
 - 12V battery with Low Voltage (LV) system power bus
 - Jumper wires for Daisy chain and Inter-Integrated Circuit (I²C) interfaces
- Test Equipment
 - STLINK-V3MINIE
 - CAN-to-Universal Serial Bus (USB) connector
 - Hair Dryer
 - Multimeter
 - Zephyr Development Environment and STM32CubeProgrammer
 - Power Supply

- Other Equipment

- Two battery modules with removable cells
- Two 24V halogen light bulbs
- Power cable connecting High Voltage (HV) system
- HV switches

D.1.3 Pre-Test Setup

1. Attach the monitoring boards to each of the battery modules. Ensure the exposed terminals of the thermistors are connected to negative terminal of the cell they're measuring.
2. Connect the SHUNT resistor LV terminals and Pack connectors to the Current SHUNT boards IN and V_{BUS} connectors respectively.
3. Assemble the rest of the HV system with the SHUNT resistor HV terminals, halogen bulb loads and switches. Only have two switches active.
4. Connect the following modules and interfaces:
 - The daisy chain connectors of the monitoring boards to each other. Then connect the south connector (COMML) of the first and the north connector (COMMH) of the last monitoring board to the COMM connectors on the BMU.
 - The I²C interface between the Current SHUNT module and the BMU.
 - The Power Lines and Alert signal to the Header next to the I²C interface on the BMU.
 - The STATUS LED on the BMU to a simple active high Light Emitting Diode (LED).
 - The 12 volt power supply and ground to the LV system power bus.
 - The CAN Fast Data (CAN) interface on the BMU to the CAN-to-USB connector.
 - The STLINK-V3MINIE with the debugging port of the BMU

D.1.4 Top Down Test Steps

1. Tap the PWR button on the BMU to activate the BMS
2. If the BMU is not pre-installed with Firmware follow these instructions
 - (a) open a terminal on the machine with the Zephyr Development Environment and STM32CubeProgrammer
 - (b) Use the Zephyr command

```
west build -p always -b nucleo_c092rc  
[Battery-Management-System-Capstone-2026 file path]/BMS_Firmware
```
 - (c) Use the Zephyr command `west flash` to flash the BMU with the `zephyr.elf` file produced from the BMS Firmware directory.
3. Read from device file connected to by CAN-to-USB connector
4. Measure the voltages of each cell, record the highest and lowest voltage, calculate the pack voltage, and compare to BMS voltage measurements.
5. Measure the voltage across the SHUNT resistor, calculate current, and compare that to the BMS pack current
6. Measure the voltage of the SHUTDOWN REQ which should be logic low or 0V, observe the LED connected to STATUS LED that should be currently off, and look at the message on the CAN-FD with id 0x010 which should be 0x00.
7. Blow hair Dryer on battery modules. SHUTDOWN REQ should be logic high or 5V, the LED is on, and the message has a value of 0x81.
8. Reset the BMS by turning off and on the BMU
9. Now remove a cell from a Battery Module. SHUTDOWN REQ should be 5V, the LED is on, and the message has a value of 0x84.

10. Attach the power supply to the empty cell terminals and reset the BMS. SHUTDOWN REQ should be 5V, the LED is on, and the message has a value of 0x88.
11. Return the normal cell to its place, reset the BMS, and add activate the third bulb switch. SHUTDOWN REQ should be 5V, the LED is on, and the message has a value of 0x82.
12. Switch off the third bulb, reset the BMS, and disconnect both the daisy chain connectors on the BMU. SHUTDOWN REQ should be 5V, the LED is on, and the message has a value of 0x90.
13. Reconnect both the daisy chain connectors, reset the BMS, and use STM32CubeProgrammer to stop the program. SHUTDOWN REQ should be 5V, the LED is on, and the message has a value of 0xA0.
14. Turn off the BMS by tapping the PWR button again

D.1.5 Post-Test Teardown

1. Remove LV power bus from BMU
2. Remove all the connections between the modules
3. Disassemble the HV system
4. Remove the Current SHUNT module from SHUNT resistor and the Monitor from each battery module.

D.2 Bottom-Up Testing

Each table below shows the test for each board.

Test Author: Jose Ramirez						
	Test Case Name:	CAN-FD Bus Test	Test ID #:		01	
	Description:	Tests CAN-FD transmission	Type:		Black Box	
Tester Information						
	Name of Tester:		Date:			
	HW/SW Version:	1.0	Time:			
	Setup:	Connect a programmed BMU with BMS_CAN Message Test to a CAN-to-USB Connector				
TEST	INPUTS	EXPECTED OUTPUTS	P A	F A	N/A	Comments
1			S	I		
2			S	L		
3						
4						
	Overall test result:					